

CS-540 Data Literacy & Visualizations

1-2 Submit Assignment: Clean Enough to Trust?

Malgorzata Debska

Southern New Hampshire University

Dr. Rahul Paul

February 3, 2025

Data Quality Evaluation of the Los Angeles Crime Data Set

Before data can be used to make informed decisions, create visualizations, or support recommendations, it is essential to evaluate its quality. High-quality data improves the accuracy and reliability of analysis, while poor data quality can lead to misleading results. This evaluates the quality of the Crime Data from 2020 to Present data set by examining its source, structure, cleanliness, and overall usefulness for analysis. In addition, the paper discusses how data quality issues may impact analysis and shows the types of questions the data set can help answer.

The selected data set, titled Crime Data from 2020 to Present, contains reported crime incidents within the City of Los Angeles beginning in 2020. The data was obtained from Data.gov, a U.S. government platform that provides public access to government-generated data. Because Data.gov is an official government website, it is considered a reputable and trustworthy source. The data is collected and kept by the Los Angeles Police Department (LAPD), which further supports its credibility. Government crime data is widely used by researchers, policymakers, and analysts, making this data set reliable for academic and analytical purposes.

The data set includes a variety of data types across its columns. Numeric data is used for fields such as crime codes, victim age, geographic coordinates, and area numbers. Categorical data appear in columns such as crime description, weapon description, premise description, and victim sex. Date and time data are used to record crimes and when they were reported. In addition, the data set has identifier fields, such as report numbers, that uniquely name each crime record. The presence of multiple data types allows for detailed analysis, including trend analysis, geographic analysis, and demographic comparisons.

Despite being well-structured, the data set has several data quality issues that could affect analysis. One of the most common issues is the presence of missing or null values. For example, some records lack victim demographic information, such as age or sex, particularly in cases where the information was not available or not reported. There are also missing values in columns related to weapons or premises, which may not apply to every crime. Additionally, some categorical fields hold inconsistent formatting, such as variations in capitalization or abbreviated values. There are also unusual entries, such as victim ages recorded as zero or extremely high values, which may show data entry errors.

Overall, the data set is relatively clean but not ready for immediate analysis without preprocessing. The structure of the data is logical, with each row standing for a single reported crime, and each column being a specific attribute of that crime. However, the missing values and inconsistent formatting require attention before analysis can be performed accurately. For example, victim age values of zero or null would need to be handled to avoid skewing averages, and categorical fields would need to be standardized to ensure correct grouping. With proper data cleaning steps, such as filtering invalid values and normalizing text fields, the data set can be considered high quality for analytical use.

If these data quality issues are not addressed, they could significantly affect the effectiveness of analysis. Missing demographic data could lead to incomplete or biased conclusions about which populations are most affected by crime. Inconsistent categorical values could result in incorrect groupings, causing inaccurate counts or comparisons. Outliers and incorrect numeric values could distort trends, making certain crime patterns appear more significant than they truly are. Addressing these issues is essential to ensure that any insights derived from the data are reliable and meaningful.

After cleaning and preparing, this data set can support a wide range of real-world questions and insights. Analysts could examine trends in crime over time, such as changes in crime frequency since 2020. The data could also be used to find which areas of Los Angeles experience higher rates of specific types of crime. Awhich areas of Los Angeles experience higher rates of specific types of crime. AAlso, the data allows for demographic analysis, such as understanding how crime affects different age groups or genders. These insights could support law enforcement planning, public policy decisions, and community safety initiatives.

In conclusion, the Crime Data from 2020 to the Present data set is a reputable and valuable resource for crime analysis within the City of Los Angeles. While the data has common quality issues such as missing values and inconsistent formatting, it is well-organized and suitable for analysis after proper cleaning. Evaluating and addressing these data quality concerns strengthens the reliability of any conclusions drawn from the data. This evaluation highlights the importance of data quality assessment as a foundational step in the data analysis process.